

Introduction to Scientific Data Management & Knowledge Graphs (Lecture 1)



The Scientific Data Management Group

Scientific Data Management Research Group



The Scientific Data Management Research Group was founded in 2021 with the goal of addressing the challenges of data management at every stage of scientific work. The research group is led by Prof. Dr. Maria-Esther Vidal.

https://studip.uni-hannover.de/sendfile.php?force_download=1&type=0&file_id=6d85f54bef48ce6a1aadfe4e0802584e&file_name=PresentationSDM.pdf

Agenda

- Organizational
- Basic Concepts
- Knowledge Representation Using Graphs
- Introduction to Knowledge Graphs

Organizational

Course pages @ STUD.IP

`https://studip.uni-hannover.de/dispatch.php/course/details?sem\_id=88e86ef9a1de34ee2134987e549753f0&again=yes`

Please, registered in StudIP:

- slides, notes, and announcements; and
- questionnaires

Online teaching (on 07.11.2025 and 19.12.2025): WebEx. For details and links, please, see the course page

Our team



Prof. Dr. Maria-Esther Vidal
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Dr. Disha
Purohit



M.Sc. Yashraj
Chudasama



M.Sc. Enrique
Iglesias



Wolfgang
Schröder

Place and Language

Lecture: Friday: 11:00 - 12:30

Location: Raum 031: e-Classroom eNIFE, Gebaeude 3702:
Informationstechnologie

Language: Lectures and Exercises will be taught in English.

Exercises, homework, and exam will be in English, and the students should answer them in English

Bonus and Exam

Bonus: To receive the bonus, at **least 9 out of the 11 homework assignments** must be submitted. The bonus corresponds to a percentage of the exam points (**EP**) calculated based on the following rules:

- If the student gets 60%-74% of **the total points of the 11 homework assignments**, then the bonus is equal to **5% of the EP**.
- If the student gets 75%-89% of **the total points of the 11 homework assignments**, then the bonus is equal to **7.5% of the EP**.
- If the student gets 90+% of **the total points of the 11 homework assignments**, then the bonus is equal to **10% of the EP**.
- Otherwise, **the bonus is 0**.

Final Exam:

- **Date:** 19.03.2026, 16:00-18:00
- **Location:** 1101.E001

Exercises

Exercises: Friday: 14:00 - 15:30

Location: Raum 010: MZ2, Gebäude 3408: Mehrzweckgebäude

Exercise page @ STUD.IP

https://studip.uni-hannover.de/dispatch.php/course/details?sem_id=f96410b9f1aa72930b75d4e102fc6c12&again=yes

Please, register in StudIP:

- Access to assignments and solutions.

Homeworks can be submitted in teams of up to two students.

SDM & KGs - Main Objectives

- Understand the main challenges of scientific data representation and integration
- Define knowledge graphs as expressive data structures to model, merge, and encode knowledge spread across heterogeneous data sources
- Study graph data models and ontologies to express and efficiently manage and store data and knowledge
- Study ontologies for describing data sources and data integration
- Study inductive learning methods on top of knowledge graphs

Literature (1/2)

- Hogan, Aidan and Blomqvist, Eva and Cochez, Michael and d'Amato, Claudia and de Melo, Gerard and Gutierrez, Claudio and Kirrane, Sabrina and Gayo, José Emilio Labra and Navigli, Roberto and Neumaier, Sebastian and Ngomo, Axel-Cyrille Ngonga and Polleres, Axel and Rashid, Sabbir M. and Rula, Anisa and Schmelzeisen, Lukas and Sequeda, Juan F. and Staab, Steffen and Zimmermann, Antoine. Knowledge Graphs. Synthesis Lectures on Data, Semantics, and Knowledge. Morgan & Claypool Publishers. 2023 <https://kgbook.org/> [6]
- Ansgar Scherp, Gerd Groener, Petr Škoda, Katja Hose, and Maria-Esther Vidal. Semantic Web: Past, Present, and Future. In Special Issue on Trends in Graph Data and Knowledge - Part 2. Transactions on Graph Data and Knowledge (TGDK), Volume 2, Issue 1, pp. 3:1-3:37, Schloss Dagstuhl – Leibniz-Zentrum für Informatik (2024) <https://doi.org/10.4230/TGDK.2.1.3>
[https://drops.dagstuhl.de/entities/document/10.4230/TGDK.2.1.3\[12\]](https://drops.dagstuhl.de/entities/document/10.4230/TGDK.2.1.3[12])

Literature (1/2)

- Gutierrez, Claudio and Sequeda, Juan F. Knowledge graphs. Communications of the ACM. Association for Computing Machinery (ACM). 2021. <https://dl.acm.org/doi/10.1145/3418294> [4]
- Mayank Kejriwal, Craig A. Knoblock and Pedro Szekely. Knowledge Graphs: Fundamentals, Techniques, and Applications. The MIT Press 2021. ISBN 9780262045094.
https://books.google.de/books/about/Knowledge_Graphs.html?id=iqvuDwAAQBAJ&redir_esc=y
- Juan Sequeda and Ora Lassila. Design and Building Enterprise Knowledge Graphs. Synthesis Lectures on Data, Semantics, and Knowledge, Morgan & Claypool Publishers 2021.
https://www.morganclaypoolpublishers.com/catalog_Orig/samples/9781636391755_sample.pdf
- Fundamentals of Database Systems Seventh Edition. Ramez Elmasri and Shamkant Navathe. (2017) [2]

Basic Concepts

- Data
- Database
- Database in Practices
- Data Management
- Scientific Data
- Challenges in Data Management
- Data and Metadata
- Data, Information, and Knowledge
- Semantics

What is Data?

Data

- facts, observations, measurements, or information
- can be analyzed, interpreted, and used to gain insights, make decisions, or solve problems.
- various forms, including numbers, text, images, audio, and video.

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Data- our new oil



<https://www.linkedin.com/pulse/data-new-oil-clark-boyd>

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- Collection of information that exists over long period of time, often many years [3].
- Collection of data that is managed by a *database management system* [3].

Databases in Practice (1/3)

- **Amazon** databases comprise more than 310 million active users globally. Amazon received over 2.72 billion unique monthly visitors in 2023. Due to accessibility and ease of use, the Amazon app has around 197 million monthly active users in 2024 ¹.
- **Facebook** has roughly three billion monthly active users as of the second quarter of 2023 ².

¹Accessed on 10.2024 <https://www.yaguara.co/amazon-statistics/>

²Accessed on 10.2024 <https://www.statista.com/statistics/264810/number-of-monthly-active-facebook-users-worldwide/>

³Accessed on 10.2024 <https://seo.ai/blog/how-many-people-use-google>

⁴Accessed on 10.2024 <https://www.searchlogistics.com/learn/statistics/netflix-statistics>

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- **Facebook** has roughly three billion monthly active users as of the second quarter of 2023 ².
- **Google** receives approximately 99,000 search queries are processed by Google every second, resulting in about 8.5 billion searches daily and roughly 2 trillion global searches annually³.
- **Netflix** has about 260 million subscribers worldwide as of the last quarter of 2023⁴.

¹ Accessed on 10.2024 <https://www.yaguara.co/amazon-statistics/>

² Accessed on 10.2024 <https://www.statista.com/statistics/264810/number-of-monthly-active-facebook-users-worldwide/>

³ Accessed on 10.2024 <https://seo.ai/blog/how-many-people-use-google>

⁴ Accessed on 10.2024

<https://www.searchlogistics.com/learn/statistics/netflix-statistics>

Databases in Practice (2/3)

DATA.nasa.gov From July 2014,
20K users and 40K datasets.



PubMed Only in 2022, 1,714,780
annual citations, 34,693,538
citations cumulative total, 2.58
Billion searches.



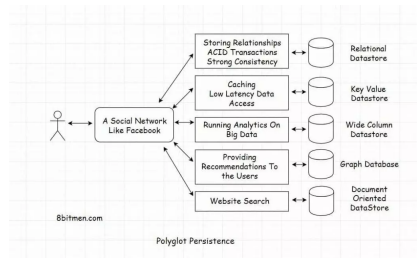
TIB Portal >95 million
bibliographic objects. AV Portal for
scientific videos with semantic
indexing >15,000 videos.



Databases in Practice (3/3)

Databases and Social Networks.

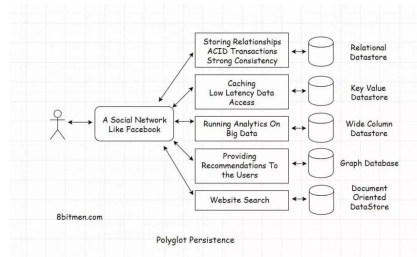
Polyglot persistence system combining different database management systems to enable efficient social network platforms.



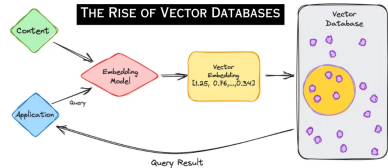
Databases in Practice (3/3)

Databases and Social Networks.

Polyglot persistence system combining different database management systems to enable efficient social network platforms.



Databases and Large Language Models (LLMs). LLMs resort to databases of vectors managed by data management systems for efficiently storing, indexing, and searching high-dimensional data points or vectors.



What is Data Management?

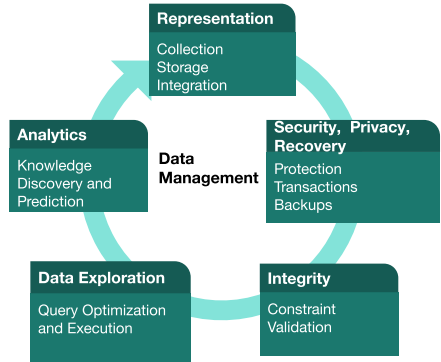
Set of Methods: for efficiently executing the following tasks over collections of data:

- Representation
- Security and Privacy Management
- Integrity Management
- Query Processing, and
- Analytics

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What is Scientific Data?

- Scientific data refers to recorded factual material commonly accepted in the scientific community as necessary to validate research findings.
- Data can take various forms depending on the discipline and context:
 - Numerical data
 - Biological samples
 - Physical collections
 - Electronic Health Records (EHRs)
- Scientific data differs from other types of materials, such as associated metadata, software, or models.
- The research data lifecycle gives context to understanding the generation, management, and use of scientific data.

Challenges in Scientific Data Management

- Scientific data sources represented in various formats:
 - Unstructured (e.g., publications and EHR) and genomic data
 - Semi-structured (e.g., short text)
 - Images
- Data sources may suffer from low data quality issues:
 - A large percentage of missing values
 - Instances that invalidate integrity constraints
- Personal data (e.g., clinical records) regulated by data protection laws Access and use require consents which can be revoked any time
- Data sharing regulations
 - Impose key principles on data sharing and processing
 - Validation and Reproducibility must be ensured

Data vs. Metadata (1/3)

Data:

- The actual content or raw facts used in analysis or decision-making.

555	Lung Cancer	Vinorelbina	3mg daily	30.10.22	01.11.2022
555	Lung Cancer	Cisplatin	10mg daily	30.10.22	01.11.2022
555	Lung Cancer	Omeprazol	75mg	30.10.22	01.11.2022

Data vs. Metadata (2/3)

Metadata:

- Data that describes or provides information about other data.

Patient

EHR	Diagnosis	Treatment	Doses	Start_Date	End_Date
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Patient data structure representing the values of the main attributes of a cancer patient.

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Patient data structure representing the values of the main attributes of a cancer patient.

- **EHR**: string attribute representing the identifier of the electronic health record of the patient.
- **Diagnosis**: string attribute representing the diagnosis of the patient.
- **Treatment**: string attribute representing the name of the medicine prescribed to the patient.
- **Doses**: string attribute representing the doses of the medicine prescribed to the patient.
- **Start_Date**: date attribute representing the date when the treatment was initiated.
- **End_Date**: date attribute representing the date when the treatment ended.

Data vs. Metadata (3/3)

Key Difference:

- Data is the actual content, whereas metadata provides context and description for that data.
- Metadata helps to understand, organize, and manage data.

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Data composed of symbols

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Symbols enriched with meaning

Data, Information, and Knowledge

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■ Knowledge:

- Insights, interpretations, and understanding derived from information.
- Example: information enriched with semantics: **Vinorelbine** is a chemotherapy drug indicated in the treatment of advanced non-small cell lung cancer.

Semantics (1/2)

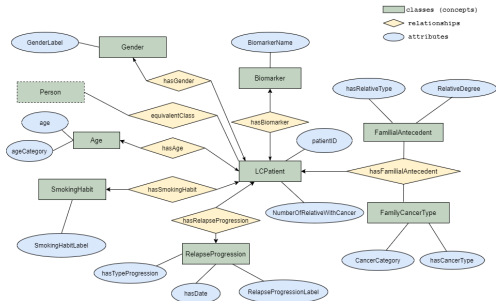
Semantics:

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- Conceptual schema defining the structure and meaning of data elements: classes, properties, and relationships.

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Semantics (2/2)

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Example of Domain Constraint:

Nivolumab is **NOT** typically used to treat patients with EGFR-positive gene mutations.

$$\forall x (EGFR_Positive(x) \rightarrow \neg Prescribed(x, Nivolumab))$$

Translation: For all patients x , if they have an EGFR-positive gene mutation, they should not be prescribed Nivolumab.

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Anomaly Detection:

A person who is EGFR-positive and appears as taking Nivolumab is flagged as an anomaly.

$$EGFR_Positive(p) \wedge Prescribed(p, Nivolumab) \rightarrow Anomaly(p)$$

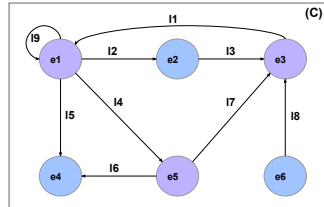
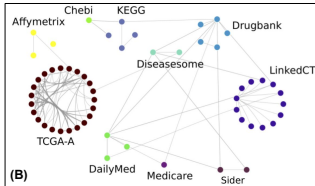
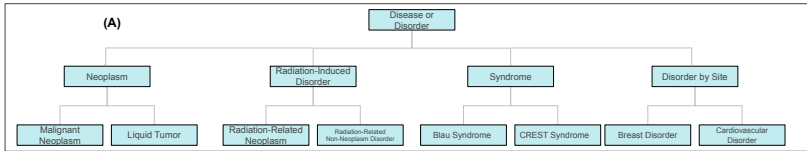
Translation: If a patient p is EGFR-positive and has been prescribed Nivolumab, then this is an anomaly in the data.

Knowledge Representation Using Graphs

- Graphs
- Ontologies
- Data Graphs
- Examples of Data Graphs

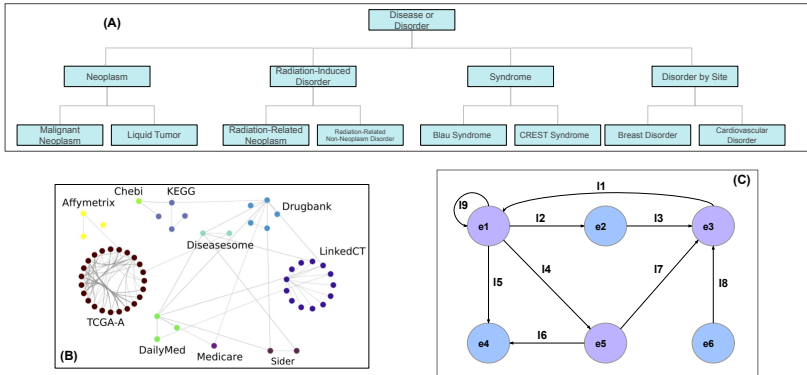
Graphs

Intuitively, a graph is a set of vertices connected by edges



Graphs

Intuitively, a graph is a set of vertices connected by edges



- (A) Tree or undirected graph in which any two vertices are connected by exactly one path.
- (B) Undirected graphs have edges that do not have a direction.
- (C) A directed labeled graph.

Ontologies (1/2)

Explicit and formal specification of a shared conceptualization:

- explicit, i.e., employs a declarative vocabulary to express the domain conceptualization
- specification, i.e., level of details of conceptualization depends on domain requirements
- language-dependent, i.e., vocabularies possess various
- expressively levels
- define domain concepts and relationships among them

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Ontologies (2/2)

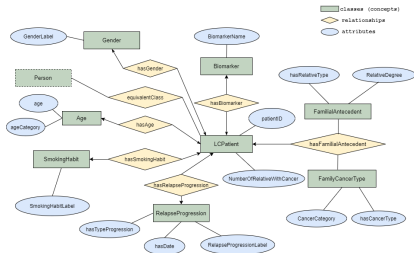
Enable to develop a common understanding of a universe of discourse in terms of:

- concepts,
- properties
- relationships
- integrity constraints

Ontologies (2/2)

Enable to develop a common understanding of a universe of discourse in terms of:

- concepts,
- properties
- relationships
- integrity constraints



Examples of Scientific Ontologies

Ontology	Description	Number of Classes
Disease Ontology (DO) https://disease-ontology.org/	Knowledge model for human diseases	> 11,000 classes
Experimental Factor Ontology (EFO) https://www.ebi.ac.uk/efo/	Knowledge model for experimental and clinical parameters	> 20,000 classes
SNOMED-CT https://www.snomed.org/	Knowledge model for clinical terminology, diseases, diagnostics, and procedures	> 300,000 classes
NCI Thesaurus (NCIt) https://ncithesaurus.nci.nih.gov/ncitbrowser/	Knowledge model to integrate molecular and cancer-related knowledge	> 110,000 classes
Ontology of Adverse Events https://www.ebi.ac.uk/ols4/ontologies/oae	Knowledge model to integrate biomedical adverse events arising from medical interventions	> 1,800 classes

Data Graphs

Data graphs are graph-based structures whose nodes represent entities and edges represent their relationships and attributes.

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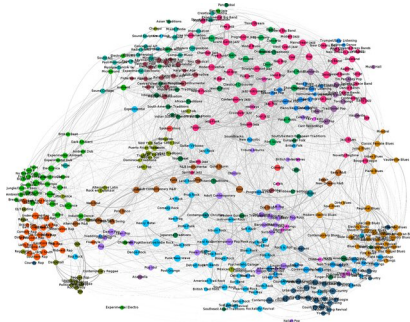
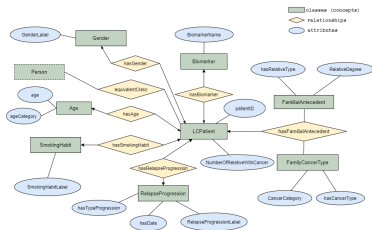


Figure: Network structure of music genres and their stylistic origin.

https://www.researchgate.net/publication/328370462_Innovation_or_deviation_The_relationship_between_boundary_crossing_and_audience_evaluation_in_the_music_field/figures?lo=1

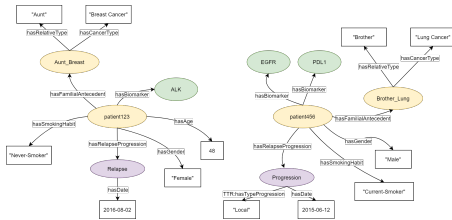
Ontologies vs. Data Graphs

Ontology



Graph-based structures representing explicit and formal specifications of a shared conceptualization.

Data Graph



Graph-based structures representing connections among data entities.

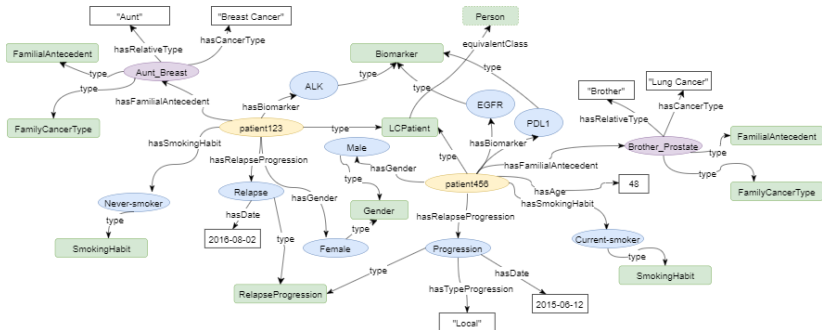
Knowledge Graphs

- Knowledge Graphs
- Key Milestones in Knowledge Graphs
- Examples of Knowledge Graphs

Knowledge Graphs (1/3)

Entities and relationships are both **first-class citizens** in a graph data model.

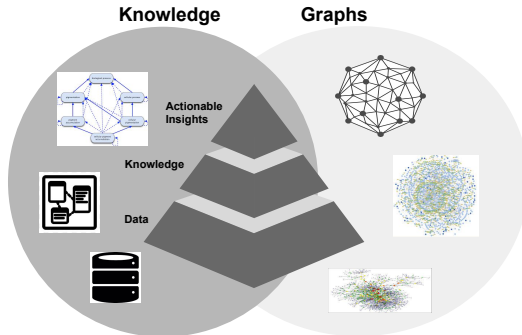
- Natural representation of a **large number of relationships** between entities.
- Enable **easy traversal** along those connections.



Knowledge Graphs (2/3)

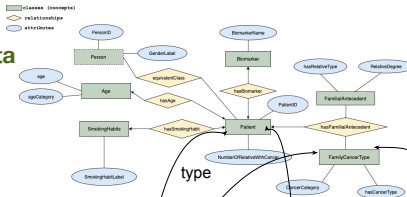
Knowledge Graphs data structures representing

- The convergence of knowledge and data as factual statements
- using a graph data model



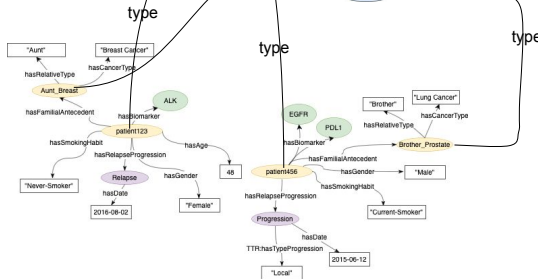
Knowledge Graphs (3/3)

Metadata

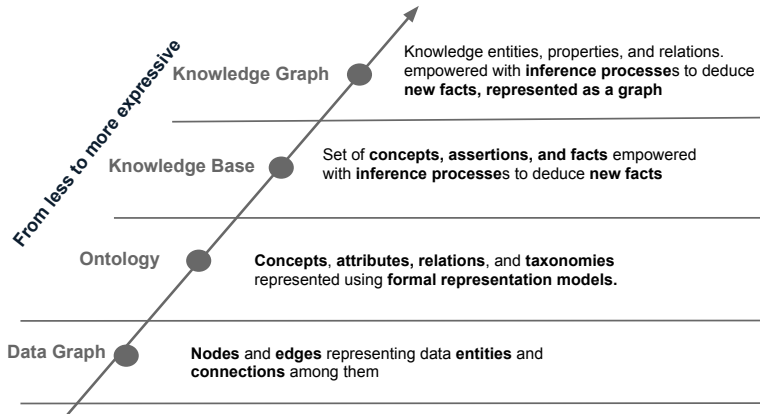


Knowledge Graphs are Graph-based data structures representing the convergence of data and knowledge (metadata)

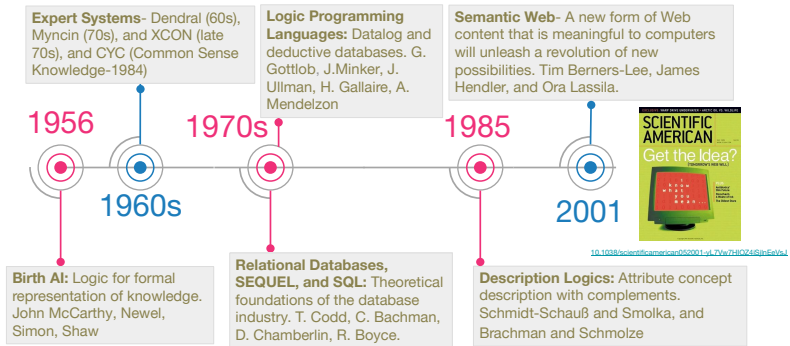
Data



Spectrum of Knowledge Representation



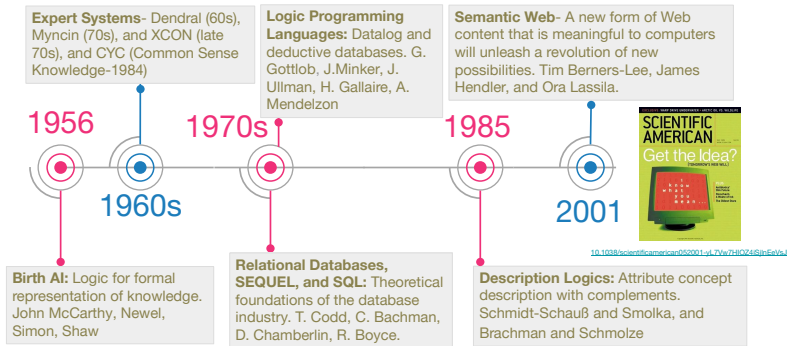
Key Milestones in Knowledge Graphs (1/2)



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⁵<https://www.scientificamerican.com/article/the-semantic-web/>

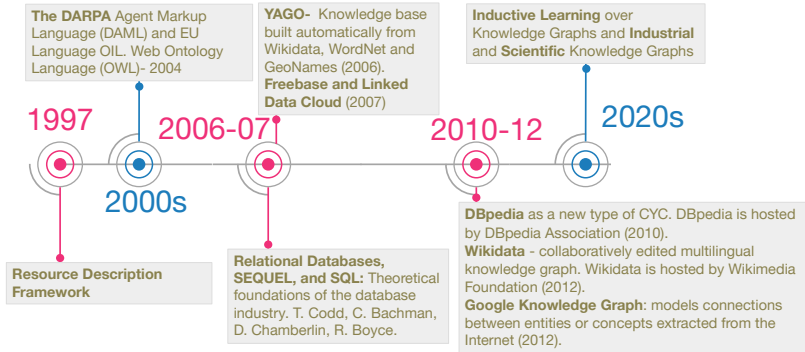
Key Milestones in Knowledge Graphs (1/2)



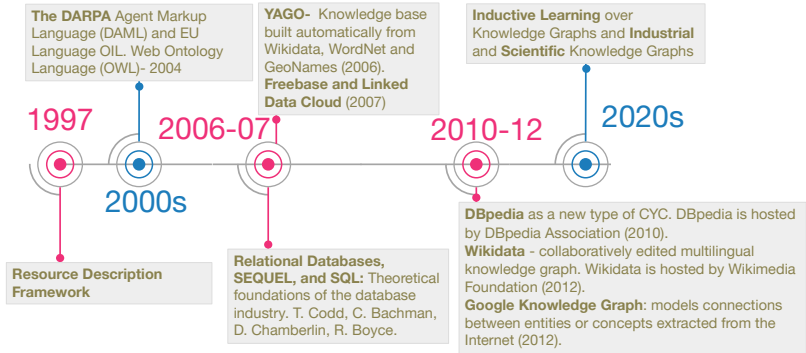
Goal: Emulate visual representation and formal reasoning ⁵

⁵<https://www.scientificamerican.com/article/the-semantic-web/>

Key Milestones in Knowledge Graphs (2/2)



Key Milestones in Knowledge Graphs (2/2)



Goal: Integrate heterogeneous data, represent knowledge using graphs, and perform formal reasoning. Knowledge graphs should be machine-understandable

Interesting Papers to Read

practice

■ Application development with **COBOL**
[continued on p. 60]

**Five diverse technology companies
show how it's done.**

**BY NATHAN NEY, YOUNG DAG, JAMES JAN,
DAVID HADZIMANIS, ALAN PETERSSON, AND JAMIE TWILVER**

**Industry-Scale Knowledge Graphs:
Lessons and Challenges**

KNOWLEDGE GRAPHS are critical to many enterprises today. This episode shows the structured data and factors

[illegible][illegible]

Semantic Web: Past, Present, and Future

Angsar Scherp 
UIN University, Germany

Gerd Groeger 
Carl-Zeiss-SMT GmbH, Oberkochen, Germany

Peter Skold 
Department of Software Engineering, Faculty of Mathematics and Physics, Charles University, Prague, Czech Republic

Katrin Hose 
TU Wien, Austria

María-Esther Vidal 
Lobata University of Hannover, Germany

TIB Leibniz Information Centre for Science and Technology, Hannover, Germany

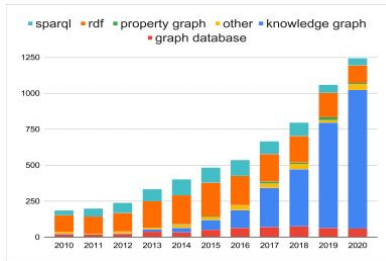
Abstract

Ever since the vision was formalized, the Semantic Web has inspired many generations of researchers. Such generations have been used to share vast amounts of information on the Web, enhance them with semantics to give them meaning, and to create and consulting on them. Throughout the years, semantic data has matured in many knowledge graphs, have been used in scientific data, data integration, enterprise settings, and creative learning.

We cover such knowledge representation, creating and validating knowledge on the Web, reasoning on it, and linking, and distributed querying. We enhance this classical view of the so-called "Semantic Web Layer Cake" with an update of recent concepts that include provenance, security and trust, as well as a discussion of practical aspects (e.g. industry) contributions. We conclude with an outlook on the future direction of the Semantic Web.

References: Bernstein et al. [1], Noy et al. [10], Sakr et al. [11], Hitzler [5], Gutierrez and Sequeda [4], and Scherp et al. [12].

Knowledge Graphs are Everywhere

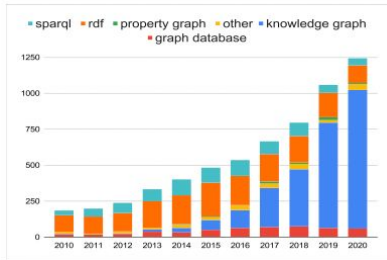


<https://marceloarenas.cl/talks/SIGMOD-21.pdf>



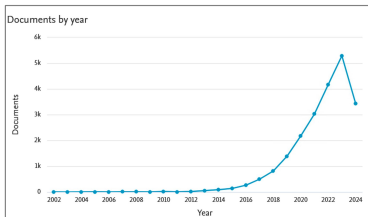
<https://kgkg.factnexus.com/@3782~167.html>

Knowledge Graphs are Everywhere



<https://marceloarenas.cl/talks/SIGMOD-21.pdf>

<https://kgkg.factnexus.com/@3782~167.html>



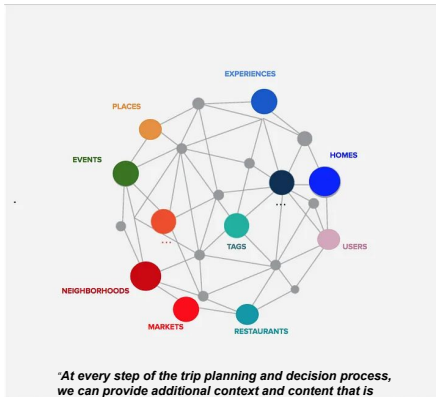
KGs: Increasingly adopted for structuring data and enhancing AI reasoning capabilities.

Scopus search 25.09.2024

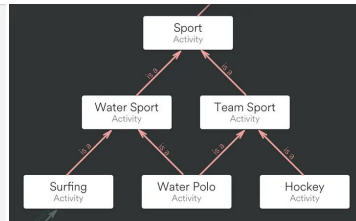
Knowledge Graphs in Industry[10]

	Data model	Size of the graph	Development stage
Microsoft	The types of entities, relations, and attributes in the graph are defined in an ontology.	~2 billion primary entities, ~55 billion facts	Actively used in products
Google	Strongly typed entities, relations with domain and range inference	1 billion entities, 70 billion assertions	Actively used in products
Facebook	All of the attributes and relations are structured and strongly typed, and optionally indexed to enable efficient retrieval, search, and traversal.	~50 million primary entities, ~500 million assertions	Actively used in products
eBay	Entities and relation, well-structured and strongly typed	Expect around 100 million products, >1 billion triples	Early stages of development and deployment
IBM	Entities and relations with evidence information associated with them.	Various sizes. Proven on scales documents >100 million, relationships >5 billion, entities >100 million	Actively used in products and by clients

The Airbnb Knowledge Graph

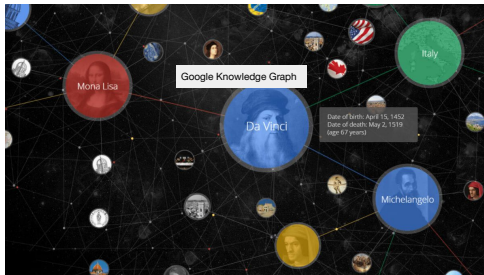


"At every step of the trip planning and decision process, we can provide additional context and content that is personalized to each individual to make it easier than ever to go from that dream trip in your head to soaking in a new environment and meeting the amazing locals."

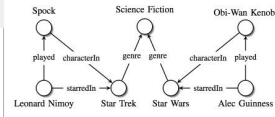


<https://medium.com/airbnb-engineering/>

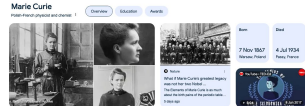
The Google Knowledge Graph



Entities and relations in the Google KG



Search on Marie Curie



Maria Salomea Skłodowska-Curie, known simply as Marie Curie, was a Polish and naturalised-French physicist and chemist who conducted pioneering research on radioactivity. [Wikipedia](#)

Born: November 7, 1867, Warsaw, Poland

Died: July 4, 1934 (age 66 years), Passy, France

Discovered: Radium, Polonium

Education: University of Paris (1903), University of Paris (1894) - [See more](#)

Spouse: Pierre Curie (m. 1895–1906)

Awards: Nobel Prize in Physics, Nobel Prize in Chemistry - [See more](#)

Children: Eve Curie, Irene Jolot-Curie

[Feedback](#)

People also search for



Nikola Tesla
Trending

Pierre Curie

Eve Curie

Albert Einstein

Related people

<https://developers.google.com/knowledge-graph>

DBpedia [8]

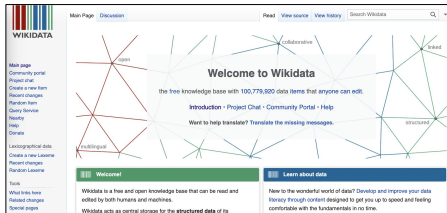


DBpedia is a knowledge graph comprising Wikipedia entities extracted from Wikipedia infoboxes:

- annotated using the DBpedia ontology
- described by short and long abstracts
- linked to external web pages about the described entities
- connected to other knowledge graphs, e.g., Wikidata, Geonames, MusicBrainz, and YAGO
- accessible via a SPARQL endpoint

<https://www.dbpedia.org/>

Wikidata [13]

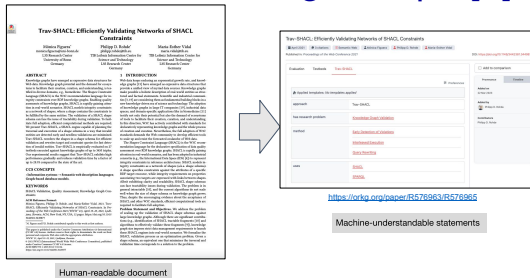


Wikidata is a Wikipedia for making publicly available structured data:

- launched on October 2012
- makes available encyclopedic knowledge as factual statements
- collaborative knowledge bases where anyone can share knowledge
- conflicting data may exist
- multilingual data with multiple formats (e.g., JSON, RDF)
- continuous evolution

https://www.wikidata.org/wiki/Wikidata:Main_Page

Open Research Knowledge Graph [7]



Open Research Knowledge Graph (ORKG) represents scholarly publications into structured, machine-readable formats.

- ORKG enables better accessibility, comparison, and reuse of scientific knowledge .
- ORKG enhances research transparency and reproducibility by providing a structured view of research outputs.

<https://orkg.org/>

Applications of Knowledge Graphs

Inductive Learning

- Knowledge Graph Embeddings
- Symbolic Rule Mining

Knowledge Fusion

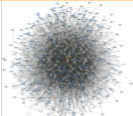
- Semantic Data Integration
- Knowledge Graph Construction
- Methodologies

Symbolic Representation

- Modeling
- Reasoning
- Consistency Checking and Verification

Knowledge Extraction

- Named Entity Recognition
- Entity Linking
- Type Membership and Resolution



Knowledge Graphs

AI Systems

- Neuro-Symbolic Approaches
- Recommender Systems
- Question-Answering
- Large Language Models

Knowledge Completion

- Link Prediction
- Entity Prediction

Scalable Management

- Knowledge Evolution
- Query processing
- Privacy and Security
- FAIR Knowledge Graphs

Applications

- Scientific Research
- Education
- Social Media
- Healthcare....

Exercise (1/2)

Search in Wikidata <https://www.wikidata.org/wiki/Q9047> and ChatGPT. Consider the questions:

- Who is Gottfried Wilhelm Leibniz?
- When and where was Gottfried Wilhelm Leibniz born?
- When and where did Gottfried Wilhelm Leibniz die?

Participate **in the survey at Stud.IP** for exploring Wikidata, while understanding factual statements. Further, participate **in the survey at Stud.IP** for consulting ChatGPT.

Are the answers from Wikidata and ChatGPT consistent?

Exercise (2/2)

Wikidata (Q9047)

Structured factual data (e.g., birth: 1646, Leipzig; death: 1716, Hanover)

Validated and referenced statements

Focus on verifiable facts: dates, places, works, affiliations

Machine-readable format (triples, IDs)

Guarantees correctness (truth-based)

ChatGPT Deduction

Narrative explanation based on general knowledge

Contextualized interpretations, may include non-verified details

Adds reasoning, historical context, and inferred relations

Natural language generation with human-like style

May hallucinate (prediction-based)

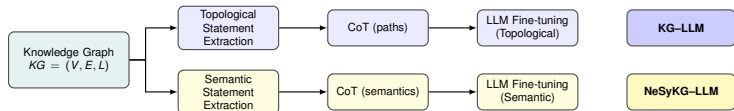
Conclusion: Wikidata provides ****validated, structured, and citable knowledge****, while ChatGPT offers ****interpreted, context-rich but fallible predictions****. Both are complementary—Wikidata ensures ***truth***, ChatGPT enhances ***understanding***.

Statement Validator

Problem. Given two entities a and b in a knowledge graph $KG = (V, E, L)$, decide whether there exists a valid, multi-hop connection $a \rightsquigarrow b$. The goal is to *validate statements* with reasoning rather than predict single links.

Approaches.

- **KG-LLM:** fine-tunes the LLM using topological paths extracted from the KG. Dong Shu et al. KG-LLM: Using Language Models to Predict Multi-Hop Links in Knowledge Graphs. ACML 2024
- **NeSyKG-LLM:** extends this by integrating semantic constraints and rules. Chudasama Y, Iglesias E., Purohit D. Vidal, M-E. NeSyKG-LLM: A Neuro-symbolic approach for to Predict Multi-Hop Links in Knowledge Graphs. 2025.



Experimental Configuration

Dataset: Synthetic Lung Cancer Knowledge Graph

KG	#triples	#entities	#relations
Lung Cancer KG	20,581	383	43

Chain-of-Thought (CoT) Generation.

Approach	#CoT samples
KG-LLM	2,501
NeSyKG-LLM	4,572

Experimental Setup.

- **Data split:** 80% CoTs for fine-tuning, 20% for testing.
- **Model:** LLaMA 3.1 (8B) fine-tuned with instruction-based prompts.
- **Metrics:**
 - **F1-score** – measures the balance between precision and recall.
 - **Accuracy** – percentage of correctly validated statements.

Experimental Results: Synthetic Lung Cancer KG

Approach	F1-score	Accuracy
LLM	0.29	0.22
KG-LLM	0.29	0.23
NeSyKG-LLM	0.68	0.83

Interpretation.

- **Topological evidence (KG-LLM)** improves structural reasoning but remains limited.
- **Semantic constraints (NeSyKG-LLM)** enable contextualized, explainable validation.
- The neuro-symbolic configuration yields the highest accuracy and interpretability.

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Conclusion. Integrating symbolic semantics from KGs allows LLMs to **reason with evidence**, transforming them into transparent and trustworthy **Statement Validators**.

Summary (1/2)

1 Data vs. Metadata

555	Lung Cancer	Vincorelbina	3mg daily	30.10.22	01.11.2022
555	Lung Cancer	Cisplatin	10mg daily	30.10.22	01.11.2022
555	Lung Cancer	Omeprazol	75mg	30.10.22	01.11.2022

EHF	Diagnosis	Treatment	Doses	Start_Date	End_Date
555	Lung Cancer	Vincorelbina	3mg daily	30.10.22	01.11.2022
555	Lung Cancer	Cisplatin	10mg daily	30.10.22	01.11.2022
555	Lung Cancer	Omeprazol	75mg	30.10.22	01.11.2022

Nivolumab is NOT typically used to treat patients with EGFR-positive gene mutations.

$\forall x (EGFR_Positive(x) \rightarrow \neg Prescribed(x, Nivolumab))$
A person who is EGFR-positive and appears as taking Nivolumab is tagged as an anomaly

$EGFR_Positive(p) \wedge Prescribed(p, Nivolumab) \rightarrow Anomaly(p)$

Symbols

Symbols with meaning

2 Knowledge Graphs

Metadata

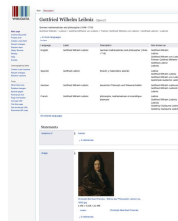
Data

Knowledge Graphs are Graph-based data structures representing the convergence of data and knowledge (metadata)

3 Various Applications of KGs

	Data model	Size of the graph	Development stage
Microsoft	The types of entities, relations, and attributes in the graph are defined in an ontology.	~2 billion primary entities, ~55 billion facts	Actively used in products
Google	Strongly typed entities, relations with domain and range inference	1 billion entities, 70 billion assertions	Actively used in products
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4 Factual Statements in Knowledge Graphs



(Gottfried Wilhelm Leibniz "occupation" mathematician)

(Gottfried Wilhelm Leibniz "occupation" jurist)

(Gottfried Wilhelm Leibniz "occupation" physicist)

(Gottfried Wilhelm Leibniz "occupation" philosopher)

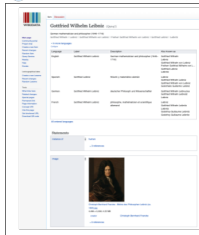
(Gottfried Wilhelm Leibniz "place of birth" Leipzig)

(Gottfried Wilhelm Leibniz "place of death" Hanover)

(Gottfried Wilhelm Leibniz "country of citizenship" Electorate of Saxony)

Summary (2/2)

4 Factual Statements in Knowledge Graphs



(Gottfried Wilhelm Leibniz "occupation" mathematician)

(Gottfried Wilhelm Leibniz "occupation" jurist)

(Gottfried Wilhelm Leibniz "occupation" physicist)

(Gottfried Wilhelm Leibniz "occupation" philosopher)

(Gottfried Wilhelm Leibniz "place of birth" Leipzig)

(Gottfried Wilhelm Leibniz "place of death" Hanover)

(Gottfried Wilhelm Leibniz "country of citizenship" Electorate of Saxony)

Factual statements:

- Correspond to labelled edges in a KG
- Can be expressed using the Semantic Web standards (e.g., RDF, OWL)
- Can be created from heterogeneous data by executing R2RML and RML mapping languages
- Can be enhanced using reasoning processes
- Can be queried using the graph-pattern based language SPARQL
- Can be validated using SHACL
- Can be discovered using inductive learning

Semester Schedule

Week	Date	Topic
1	17.10.2025	Introduction KGs
2	24.10.2025	Data Graphs and RDF
3	07.11.2025	KGs and Standards for Query Processing (Online class)
4	14.11.2025	Semantic Data Modeling
5	21.11.2025	Knowledge Representation (RDFS and OWL)
6	28.11.2025	Knowledge Representation (RDFS and OWL)
7	05.12.2025	Knowledge Graph Representation (OWL)
8	12.12.2025	KG Creation & Enrichment
9	19.12.2025	Integrity Constraints and Validations (Online class)
10	09.01.2026	Integrity Validations and KGs in Practice
11	16.01.2026	KGs in Practice
12	23.01.2026	Symbolic Learning and KG Embeddings
13	30.01.2026	KGs, LLMs, and Future Directions

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